

STATEMENT OF WORK
FOR
ARCHITECT & ENGINEER SERVICES (A&E) FY27
TO SUPPORT 86 CIVIL ENGINEER SQUADRON

09 April 2026

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1 GENERAL INFORMATION

The Architect & Engineer Services (A&E) shall provide professional architectural and engineering support services for projects assigned to the 86th Civil Engineer Squadron. The A&E shall furnish all required personnel, management, facilities, supplies, equipment, instruments, transportation, and materials (other than those furnished by the Government as provided herein or in individual task orders), as well as any other items and services necessary to accomplish the required work. The A&E shall also provide the Government with all submittals/ documents supporting material developed during the period of performance. During execution of the work, the A&E shall provide adequate professional supervision and quality control to ensure the accuracy, quality, completeness, and timely progress of all services. The general A&E services will primarily support the Kaiserslautern Military Community (KMC) area.

1.1 LOCATION OF THE PROJECT

Contracts will be awarded for work that will involve the design of various U.S. military projects located within the KMC area. However, the Government may also require work to be performed in other locations within the United States European Command.

1.2 SCOPE OF WORK

This Statement of Work (SOW) defines the general requirements for the performance of various services under this contract. Multi-disciplined A&E services are required for the design of various U.S. military buildings, structures, and facilities throughout Europe. The HOAI (Honorarordnung für Architekten und Ingenieure) serves as a binding framework for the provision of all necessary services to produce a design that is eligible for approval and ready for construction. Firms must use the latest version available for the HOAI located at <https://www.hoai.de/hoai/volltext>. All work shall be carried out in compliance with the general rules of the trade. If regulations of the U.S. Forces are more stringent than German standards (e.g. DIN, VDE, VDI, DVGW) and regulations (e.g. LBauO RLP), the U.S. regulations (UFC, UFGS, Military Standards, and the Whole Building Design Guide) shall apply. Additionally, the basis for design shall utilize all applicable guidance, including but not limited to UFC, UFGS, Military Standards, and the Whole Building Design Guide, available at <https://www.wbdg.org/dod>. If additional special services are required to achieve these objectives, the contractor shall identify and price these services in the proposal. Specific disciplines required are, but not limited to, as follows:

- Project Management
- Architectural Services
- Cost Estimator
- Interior Design and Space Planning
- Landscape Architecture
- Electrical Engineering
- Communications Engineering
- Cybersecurity Engineering
- Civil Engineering
- Structural Engineering
- Mechanical Engineering

- Fire Protection Engineering
- Environmental Engineering
- Security Engineering
- SCIF Planning, Design and Construction /Intelligence Community Directive ICD 705 subject matter expert
- Construction Site Security Planning (CSSP)/Security Manager Services (SSM)
- Geotechnical
- Surveying
- High Altitude Electromagnetic Pulse (HEMP) subject matter expert

Design work involves, but is not limited to major and minor new construction, and alteration, repair or rehabilitation of various structures such as: barracks, multi-family and single-family housing units, utility systems, vehicle maintenance facilities, administrative buildings, secure facility, above and underground storage tanks (AST and UST), gas stations, oil-water separators (OWS), Grease traps (GTs), pavement (streets, parking lots), Airfield (Runways, Taxiways, Ramps), storm and sanitary sewer, Rain Water Retention basins, Rain Water Overflow basins, water and other U.S. military facilities.

A&E services may be required for developing Request-for-Proposal (RFP) packages for design-bid-build projects. Projects may involve security upgrades, feasibility studies, and construction supervision and inspection.

A&E services may include: charrettes, engineering assessments and systems analysis of existing facilities and infrastructure, anti-terrorism/force protection (ATFP) assessments and design, site investigations and technical assessments, structural analysis and testing, feasibility studies (to include cost estimating, energy conservation, fire protection, electrical, HVAC, storm and sanitary sewer, water, pavements, environmental assessments, etc.), environmental assessments and technical assistance, occupational safety and health coordination, contaminant surveys (asbestos, mineral hydrocarbons, etc.) and abatement design, energy conservation and fire protection, minor project scope development, cost estimating and analysis, project schedule development and network analysis, construction management, inspection and quality assurance, safety inspections, materials and equipment testing including systems commissioning, technical review of designs and cost proposals, bid ability, constructability, operability and environmental compliance reviews, value engineering studies and other general engineering and design services.

NOTE: All submittal documents (e.g., specifications, drawings, design analyses, reports, studies, etc.) and any other official correspondence shall be provided in English (USA) and German. In the event of a conflict, English (USA) shall take precedence. AI translation is not permitted.

All contractors and their subcontractors are required to maintain all necessary certificates and licenses in compliance with U.S. and Host Nation laws and regulations during the entire period of performance.

All support and design services shall include associated travel, reproduction, and compliance with

U.S. and Host Nation local laws, regulations, construction practices, and procedures applicable to the location where the work/project is performed.

1.2.1 The A&E shall perform all or part of the following services as more specifically identified in task orders:

- General services include, but are not limited to, the design of various repair, rehabilitation, maintenance and minor construction projects.
- General services and ABG75/ABG2 direct procedure coordination. The A&E shall provide all required documentation and in-person coordination and shall update the design as necessary based on requirements issued by the German government during this process. The A&E shall also provide all Host Nation approval permits and documentation in the final submittal.

1.2.2 Package Submittals (e.g., reports, charrettes, studies, drawings, specifications, design analysis, cost estimation, etc.), unless otherwise stated in the Task Order (TO) SOW, design submittals shall be at 15% (Concept), 35% (Preliminary), 95% (Pre-Final) and 100% (Final) design. The A&E shall furnish minimum three (3) copies, including digital CD, unless otherwise specified in the TO. Drawings shall be provided in the latest AutoCAD version as well as in Adobe Acrobat (PDF) format. All drawings must include a consolidated list of plans and must be created and labeled in accordance with the Ramstein AB standard. Electronic file names shall comply with Ramstein AB naming conventions. Standards and naming conventions will be provided with the TO. All other submittals shall be in the latest MS Office format and as PDF files.

1.2.3 Site investigations carried out by A&E shall identify all special conditions and dimensions that may affect or be affected by the proposed work. Whether for a new project or renovation, all field investigations shall be thorough and detailed to ensure an accurate representation of existing conditions. The A&E shall not rely solely on existing information such as utility drawings or record drawings for the new design but shall verify all critical existing information.

1.2.4 The A&E shall document each site visit and meeting within ten (10) business days. The written records shall include subject, contract and TO#s, names of participants, and a summary of discussions.

1.2.5 Hazardous Material. The A&E will be responsible for securing samples and testing (by a qualified laboratory) of any suspicious materials. Samples may be required on suspect building materials, architectural finishes, equipment, piping, debris, and dust. The A&E shall submit a report identifying the number of samples taken, the location of each sample, and the corresponding test results. A summary of locations, material types, and quantities shall be included in the construction documents.

1.3 **A&E RESPONSIBILITIES AFTER TASK ORDER COMPLETION**

During the construction project solicitation and/or construction period, if discrepancies, oversights,

or omissions become evident, the A&E shall correct the affected drawings and specifications at no additional cost to the Government. Corrected drawings and/or revised documents shall be submitted to the Contracting Officer for review and approval. If needed, sample revisions and amendments will be provided to the A&E as a guide for use in preparing amendments and revisions.

Upon notification of a discrepancy, oversight, or omission, the A&E shall have up to ten (10) business days to prepare and submit an amendment or revision. The A&E shall provide engineering advice during the project construction period in connection with the design, as may be requested by the contracting Officer.

- 1.3.1 Request for Information. During solicitation and/or construction, requests for information (RFI) may occur from the construction contractor regarding the A&E services. When required, the Contracting Officer will forward the RFI to the A&E for response. The A&E shall provide answers to RFI within two (2) business days of receipt of the RFI at no additional cost to the Government.
- 1.3.2 Errors and Negligent Performance. The A&E shall be responsible for the professional quality, technical accuracy, and the coordination of all designs, drawings, specifications, and other services furnished by the A&E. As such, the A&E shall correct or revise any errors, deficiencies in its designs, drawings, specifications, and other services furnished under the A&E contract at no additional cost to the Government.
- 1.3.3 Design errors or omissions resulting in damages or extra cost to the Government will be evaluated for potential A&E financial liability. If the Government determines the A&E is financially liable for a design/construction deficiency, the A&E will be advised by official correspondence.

1.4 **TASK ORDER REQUIREMENTS**

- 1.4.1 Release and Use of Information. The A&E shall refer all inquiries concerning the contract to the CO. Under no circumstances shall any statement be released to the news media directly by the A&E or any of the employees of the A&E. Information developed, gathered, or assembled in fulfillment of the contract requirements, as defined in or related to the SOW shall not be released by the A&E, its consultants, subcontractors or their associates without prior coordination and approval by the CO. Upon completion of work, all such information shall become Government property and shall not be used by the A&E for any purpose without the written consent of the CO. The A&E shall take all necessary precautionary measures to maintain the confidential nature of all work prepared under this contract. The obligation to maintain confidentiality shall extend beyond the completion of any Task Order.
- 1.4.2 Laws and Regulations. The A&E shall perform all studies and design efforts in full compliance with applicable U.S. and Host Nation (HN) laws and regulations, as well as the country-specific Final Governing Standards. The A&E shall also comply with all

Occupational Safety and Health Administration (OSHA), Environmental Protection Agency (EPA), NFPA, USAFE, and HN federal and local safety requirements. In the event of a conflict between these regulations, the most protective standard shall apply. The A&E shall be responsible for ensuring that its agents, employees, and subcontractors perform all work in a safe manner. The A&E shall ensure that all personnel involved in soil and groundwater sampling are trained in appropriate field procedures and first aid. The A&E, its subcontractors, employees, and agents, shall possess all required certificates and licenses necessary for performance of the work.

- 1.4.3 A&E Responsibilities. Whenever possible, the Government will furnish maps showing the location of the proposed work. When borings are required, the Government will provide assistance in obtaining utility clearances. All other aspects of the contract are the A&E's responsibility. The A&E understands that any offers of assistance or work directives whether verbal or written by Government personnel other than the Contracting Officer (CO) are unauthorized, and the A&E shall not accept or act upon such offers.
- 1.4.4 Documented Information. The A&E shall provide a record of all meetings, conferences, discussions, or verbal directives that are participated in by the A&E or its representatives. These records shall be dated and shall identify participating personnel, subjects discussed and conclusions reached. The A&E shall forward to the Authorized Government Point of Contact a copy of the memorandum of the meetings within ten (10) calendar days of the occurrence.
- 1.4.5 Annotated Review Comments. The A&E shall provide written responses to all review comments and deliverables provided by the Government within ten (10) business days after receipt of review comments. The responses shall clearly reference the comments made and the specific paragraph referenced. The A&E shall also indicate what type of action they will be taking as a result of the comment. This may consist of acceptance of the comment and incorporation of the recommended change or need to further discuss to resolve differences.
- 1.4.6 Quality Control. Submittal documents and reports shall be reviewed by the A&E before being submitted. The A&E shall coordinate the design of all disciplines and shall perform a quality review to ensure that the project is buildable and that no conflicts exist between the project drawings and specifications. These reviews shall be performed by qualified personnel other than those who prepared the documents. The A&E is responsible for the completeness and accuracy of all work performed under this contract, and for compliance with all requirements of the contract, task order, and for properly identifying the correct design and/or study conclusions in accordance with all applicable laws and regulations. The Government will rely upon the professional quality of the work performed by the A&E.

1.5 GEOSPATIAL DATA DELIVERABLES

This section defines the contract requirements for a comprehensive range of geospatial data for all projects awarded under the TO. The contractor shall perform all work necessary to capture,

create, process, manage, and deliver geospatial data in support of various projects. The use of the term “data” is inclusive of tabular, vector, and raster data types.

- 1.5.1 Geospatial General Requirements. The contractor shall provide all personnel, equipment, tools, materials, supervision, and all other items and services necessary to perform the geospatial data services as defined herein. All work shall comply with the standards and requirements.
- 1.5.2 Geospatial Data Capture, Collection, Creation, and Maintenance. The contractor shall capture, collect, create, and process geospatial vector and raster data. At times the contractor will need to create land survey control markers/monuments.
- 1.5.3 In-the-Field Collection. Perform field collection and validation of digital vector data using GPS technologies (survey, resource, or mapping grade), traditional land surveying technologies (e.g., Total Station), and remote sensing technologies. All data collected using land surveying tools shall be in accordance with current industry-acceptable surveying protocols. Including the collection of associated feature attribute data and metadata.
- 1.5.4 Land Survey Control Markers/Monuments. All data collected using land surveying tools shall be in accordance with current industry-acceptable surveying protocols.
- 1.5.5 Data Quality Assurance and Quality Control (QA/QC). The contractor shall provide data QA/QC support using industry-leading tools and techniques, addressing data geometry, topology, attribution, and metadata. This includes Performing data quality assessments and delivering detailed reports and ensuring all government-owned data is compliant with mandated standards for accuracy and completeness.
- 1.5.6 Data and Deliverables Standards. All contract deliverables shall adhere to all standards in Section 1.5 of this document. The specific standards applicable to each project will be confirmed at the Task Order level by the designated government office (e.g., Ramstein AB GeoBase office).

Contract deliverables include, but not limited to:

- Site plans
- Record drawings
- Engineering designs, plans or surveys
- Topographic surveys or studies
- Boundary or Cadastral surveys
- Master Plan drawings
- Utility (water, sewer, power, storm, etc.) designs, plans, surveys and studies
- Pavement, Grading, or Excavation plans
- Soil/Geology studies or surveys

- Environmental assessments, surveys, studies, or plans
- Historical or Archaeological surveys, studies, or plans
- As-Builts

All file names and text content in the geospatial data shall be in English. The only exception to this requirement is for German place names. All final geospatial data must be delivered as an ESRI File Geodatabase that is compliant with the Air Force's Spatial Data Standards for Facilities, Infrastructure, and Environment (AF SDSFIE).

- 1.5.7 Data Schema and Standards. All data shall be compliant with the latest version of the applicable spatial data standard, such as the AF SDSFIE. Each new Task Order will need to request an updated database from the GeoBase office. Data must conform to the schema and requirements outlined in data standards and required fields to be filled out.
- 1.5.8 Data Format. All vector data deliverables shall be provided in the current ESRI File Geodatabase (.gdb) format. Shapefiles (.shp) are not an acceptable primary deliverable format. Files natively created as shapefiles and then exported to a geodatabase will not be accepted. All raster data shall be provided in a current, ESRI-compatible format.
- 1.5.9 Projection and Coordinate System. All geospatial data must be delivered in the following coordinate system, unless otherwise specified in the Task Order:
- Coordinate System: GCS_WGS_1984
 - Projection: Transverse Mercator
 - Zone: UTM Zone 32N
 - Horizontal Datum: World Geodetic System 1984 (WGS84)
 - Vertical Datum: Mean Sea Level (MSL) via Earth Gravitational Model 2008 (EGM2008)
 - Units: Meters
- 1.5.10 Data Accuracy, Positional Accuracy. All data except for utility data must be collected with sub-meter accuracy. Utility Underground Data: Must be collected at survey-grade accuracy of 1 foot or less. Data must be obtained at the time of construction and prior to burial.
- 1.5.11 Data Attribution. The GeoBase office will define the required attribution prior to the release of the task order so the contractors can properly bid on the level of effort. Any contract modifications will require a review of the attribute requirements and included in all contract modifications. The contractor shall populate all required attribute fields as defined in the applicable data standard for each feature. For data related to real property, attributes (RPUID, RPSUID, CATCODE, etc.) must match the associated DD Form 1354.
- 1.5.12 Require Attributes for Key Features, Buildings and Structures, Utilities. Measurement fields, Measurement, RPSUID, GEOLOC, Create Date, Creator (Company Name), Install Date, Material Type, Use (Utility Network Subtype), Condition Rating, and Operational

Status. Contractor can reach out to Ramstein AB GeoBase office or AFCEC GIS Utility team (AFEC.UtilitiesGIS@us.af.mil).

1.5.13 Topology. The contractor shall utilize topology rules in current ESRI software to ensure data is topologically sound and geometrically correct. Data must be free of overshoots, undershoots, dangles, slivers, null geometries, and duplicate features. Polygons must be closed and must not overlap within the same layer. Exceptions to topology rules require justification in the data documentation.

1.5.14 Deliverables and Submission. The contractor shall submit the following for all work performed, as applicable:

- Geospatial Database. A geo-referenced current ESRI File Geodatabase (.gdb) containing all features and data collected or created.
- Maps/As-Builts. Digital site plans, record drawings, and other map products as required.
- Quality Control (QC) Report. A report detailing the QC process, the software used, and a statement detailing any corrected inconsistencies or remaining errors.
- Supporting Files. All symbol/font libraries, color/pen tables, and plot configuration files necessary for a complete project.
- File names and supporting files shall be in English.
- All deliverables must be submitted as per section 1.2.2 of this document.

1.5.15 Data Review and Acceptance. All submitted data is subject to review and approval by the Government, including the Installation GeoBase Manager and relevant Subject Matter Experts (SMEs). For the 35% submittal, the GeoBase team will have a maximum of ten (10) working days to provide review and input. No data will be considered final or accepted without a formal QA/QC review and written acceptance from the designated government representative. The Government reserves the right to reject any deliverables that do not meet the standards outlined in this SOW and the corresponding Task Order. The contractor shall be entitled to payment for accepted deliverables only. The contractor will be liable for all costs and delays incurred with unacceptable deliverables.

2 A&E SERVICES:

The A&E shall perform all work required to accomplish the intent of each Task Order and, unless otherwise specifically stated therein, shall provide all required services in accordance with the following requirements.

2.1 STUDIES AND INSPECTIONS

The A&E shall perform all field and office work as required to accomplish specific studies.

2.1.1 Reports. The study shall include a discussion of the investigation, a description of all plans and schemes considered for achieving the desired end result, all drawings and sketches necessary to present the study's details and findings, and the required construction cost

estimates, including comparative costs for any alternate plans or schemes. The study shall also include results, conclusions, and recommendations. All reports shall be spiral-bound.

- 2.1.2 Drawings and Sketches. Drawings and sketches shall be prepared as required to present the details and results of the study. All drawings and sketches shall be produced at an appropriate scale in the approved AutoCAD format and shall be folded, if requested.
- 2.1.3 Cost Estimates. Required cost estimate shall be prepared for all indicated components or features of the project.

2.2 PLANNING CHARRETTE REPORT (PCR)

The A&E Team shall conduct the Charrette which will be a collaborative effort by the design team, user and client representatives, facility engineering personnel, and other stakeholders. The purpose of the Design Charrette is to validate and clarify the project requirements. The minimum list of deliverables the A&E should provide is listed below.

- 2.2.1 Draft DD Form 1391. Provide a draft DD Form 1391 and all supporting materials that define the detailed scope of the project. The submittal shall include a cost estimate developed using parametric (PCE) methods based on thorough site investigations, engineering analysis, and stakeholder interviews. The draft DD Form 1391 and associated cost estimate shall be prepared using PAX as the authorized software.
- 2.2.2 Draft Economic Analysis
- 2.2.3 Draft Simplified Analyst of Alternatives. The alternatives to construction may include lease, renovation, addition, alteration, use of existing facilities, new construction, or other feasible options.
- 2.2.4 Life cycle cost for each of the Simplified Analyst of Alternatives as described in 2.2.3
- 2.2.5 New Construction. Provide the rationale for selecting new construction as the preferred alternative, if applicable, to demonstrate why it best satisfies the project requirements.
- 2.2.6 Draft Parametric Cost Estimate (PCE). Provide a Draft Parametric Cost Estimate (PCE). The PCE is the budget-level cost estimate developed through parametric methods and feeds into the DD Form 1391. The PCE shall use parametric unit costs based on the type of construction and the quality of materials, in accordance with the DoD Guidance Unit Cost (DoD GUC). The PCE shall also include all required reports that verify and support the final scope definition. The PCE shall be prepared using PACES as the authorized software.

2.3 CONCEPT DESIGN (15%)

This scope covers HOAI phase 1 and 2. If the TO requires an ABG2 procedure, the contractor shall submit "Pendelliste" form identifying all personnel and subcontractor involved in the ABG2 process, along with proof of their certifications. This submittal shall occur at the very beginning of the Concept Design phase.

The minimum list of deliverables the A&E should provide is listed below:

- 2.3.1 Lifecycle Cost Analysis. Prepare a Lifecycle Cost Analysis for each COA identified at the Design Charrette if applicable.
- 2.3.2 Analysis of Alternatives. Provide a simplified Analysis of Alternatives for each previously identified COA.
- 2.3.3 Concept Floor Plans. New work and demolition, with room names and basic dimensions.
- 2.3.4 Concept Building Elevations with exterior materials indicated.
- 2.3.5 Concept Building Sections with heights of critical elements indicated.
- 2.3.6 Concept cost estimate
- 2.3.7 Concept explanatory report
- 2.3.8 Concept Site Plans. Indicate above and below grade utilities, vehicle and pedestrian paths, buildings, AT/FP standoff distances, parking areas, landscape concepts and existing site features that are to remain

2.4 PRELIMINARY DESIGN (35%)

This scope covers HOAI phase 3 and 4. The Preliminary Design Submittal is intended to convey the complete extent of the work in a preliminary manner. The general direction and details of the design shall be firmly established and clearly indicated at this design level. Drawings shall include basic floor plans; preliminary architectural details; mechanical and electrical equipment and distribution layouts; schedules; tables; and any other information required to clearly illustrate the design. If the TO requires an ABG2 procedure, all documents necessary for the notification process with the German authorities shall be submitted. The specific documentation requirements for each TO are define based on the "Pendelliste".

- 2.4.1 Preliminary Design Submittals will be identified at the TO level, and shall include the following, at a minimum:
- 2.4.2 Test and survey results, complete design calculations, product and equipment data, and any other information required to clearly illustrate and describe the design.
- 2.4.3 Drawings. Preliminary designs shall be submitted as scale line drawings prepared at contract plan scale as required in the task order SOW and in sufficient detail to show adequacy of design, the utilization of the item involved, and conformance of the design with the intent of the project as set forth in the instructions.
- 2.4.4 DIN 276. Preliminary cost estimates shall be based on contract drawings completed to the percentage established in the TO and other designated data. The estimates shall be accurate and realistic, reflecting current prices for work and materials. Each major item shall further be broken down into component units of labor and materials with the quantity

and cost of each unit shown. Lump sum amounts for major items that cannot be readily analyzed will not be accepted. Preliminary cost estimates shall be organized to cover the format and will represent an accurate cost for each applicable division of work.

- 2.4.5 Construction Cost Estimate. Provide construction cost estimate that complies with the current Ramstein AB Multiple Award Construction Contract (MACC) procurement process. If applicable, separate costs for new construction and renovation. In addition, provide price schedule without prices. The cost estimates shall be accurate and realistic, reflecting prices for work and materials of construction. The estimates shall be broken down into detailed items of major cost groups. Each major cost group shall be further broken down into items including both labor and material with units, quantities and costs of each item.
- 2.4.6 Explanatory Report. Prepare and submit explanatory reports for all original designs prepared by the A&E, including a concise description of the building. Complex issues shall be explained as necessary. Alternative studies and their results shall be documented in the explanatory report for all specialized planning disciplines. The report shall also describe the building site and its development. The design description shall include information on the design concept, design requirements, exterior design, modules, grid system, installation routing, adaptability and future use, expansion potential, and the application of sustainable construction principles. It shall also reference any special external conditions that significantly influence the design and affect both construction and operating costs. The description of the building structure, technical systems, and outdoor facilities shall include details regarding their construction, including materials and key properties. This description shall be supplemented with justification for the selected construction methods and materials, particularly where these choices have a substantial impact on construction and operating costs.

2.5 Pre-Final (95%) / Final Design (100%)

This scope covers HOAI phase 5 and 6. The Pre-Final Design submittal shall provide a complete and fully coordinated set of drawings, DIN276, cost estimate, design analysis and specifications. The purpose of this submittal is to allow the Government to review and approve the pre-final documents. Once Government approval is granted, the A&E shall proceed with the Final Design Submittal as outlined below:

- 2.5.1 Accomplish the final detailed design of the project, complete with all component parts and appurtenances. The final design shall be based on the established project criteria and, where furnished or required, on the approved preliminary design. The final design shall remain within the target construction cost and shall be completed in accordance with the approved cost estimate, as applicable, and within the approved project funding.
- 2.5.2 For structures requiring site adaptation of existing plans, the final design shall include all modifications, additions, and appurtenances necessary to adapt the design to local site conditions. The A&E shall ensure that all such adaptations are fully coordinated and

integrated into the final design documents.

- 2.5.3 Design Analysis. Prepare and submit a comprehensive Design Analysis addressing all engineering disciplines. The document shall serve as the complete basis of design and shall include statements of all criteria, loads, factors, assumptions, and codes used in the final design. The Design Analysis shall also incorporate all supporting diagrams, charts, and calculations developed to substantiate the design.
- 2.5.4 Drawings. Prepare complete, detailed drawings for use in the construction contract for the project. Final drawings shall be based on approved designs and layouts and shall be developed on sheet sizes specified in the Task Order SOW. The drawings shall show coordinates or ties to base lines for at least two adjacent corners of each building and structure, as well as principal points on utility lines, roads, railroads, pavements, and other major construction features. Contract drawings shall be prepared for the entire project designed under this contract and shall support the solicitation of bids and the award of construction contracts in accordance with the contract specifications. All drawings shall be prepared in an AutoCAD format that is 100 percent compatible with Ramstein AB standards, or as otherwise approved by the Contracting Officer (CO). Submittals shall include both AutoCAD files and Adobe Acrobat (pdf) files, provided on a CD along with the printed documents.
- 2.5.5 Specification (UFGS) / Leistungsverzeichnis (LV) in accordance with VOB/A. Specifications shall be concise, definitive, and free of ambiguities or omissions that could result in disputes or contractor claims for additional compensation. Specifications shall include detailed requirements for all materials and equipment. All specifications, including technical and special conditions, shall be prepared for the entire project in a single volume covering all items of work. This volume shall include the complete Statement of Work (SOW), Special Conditions, Technical Provisions, and a list of Bid Items and Quantities for the project. The Technical Provisions shall consist of a master technical specification covering all work common to the line items included under the task order. They shall be organized into separate sections corresponding to the various trades and work classifications. Final specifications, including all technical provisions, shall be submitted in the format required by the Task Order SOW or in a format otherwise acceptable to the TO.
- NOTE:** The A&E is responsible for identifying all applicable U.S. (e.g., UFC, UFGS, MIL-STD, etc.) specification requirements and ensuring they are incorporated into the project specifications in the LV format as required.
- 2.5.6 DIN 276 Calculation of Cost. Final cost estimates shall be based upon the completed contract drawings and specifications. This estimate comprises a revision of the preliminary cost estimate.
- 2.5.7 Construction Cost Estimate. Provide construction cost estimate that complies with the

Ramstein AB procurement process. The Ramstein AB Cost standard shall be used for this purpose. If applicable, separate costs for new construction and renovation. In addition, provide price schedule without prices. The cost estimates shall be accurate and realistic, reflecting prices for work and materials of construction. The estimates shall be broken down into detailed items of major cost groups, as defined in the Ramstein AB Cost Standard. Each major cost group shall be further broken down into items including both labor and material with units, quantities and costs of each item.

- 2.5.8 Submittal List. The A&E shall prepare a submittal list identifying all materials and construction items that require sampling and approval by the contractor prior to installation. Air Force Form 66 shall be used for this purpose.
- 2.5.9 The A&E shall prepare and submit the construction schedule and associated construction phasing plan. The submittal shall clearly define the sequence of work, major milestones, critical path activities, and any required phasing necessary to maintain operations, ensure safety, or meet project constraints.

2.6 **SPECIAL SERVICES**

- 2.6.1 Quality Assurance Phase (Title II Services). This scope covers HOAI phase 8. Services related to a specific construction project and consisting of supervision and inspection of construction. Quality Assurance services may comprise all aspects of construction oversight and quality assurance. This may also include attending site visits, reviewing material submittals, daily inspection report, performing periodic site inspections and preparing change orders. Quality Assurance services shall be performed by individuals who possess, at least, a degree in Engineering, Construction Management, or applicable field related to the overall project tasking. The proposed lead on-site Quality Assurance inspector shall have a minimum of 5-7 years of experience in construction practices, procedures, and testing, to include three (3) years of quality control management of DoD construction projects. Quality Assurance services may be required during non-standard duty hours, weekends, and Government holidays to support project completion requirements.
- 2.6.2 Real Property Documentation. The Contractor shall prepare Real Property Data for each TO utilizing the DD Form 1354. A draft DD Form 1354 and the "Building Information Checklist" shall be developed IAW Unified Facilities Criteria (UFC) 1-300-08.
- 2.6.3 Fire Protection. The A&E shall have on staff, or shall retain, the services of a registered Fire Protection Engineer and/or Brandschutzingenieur who meets the qualifications and service requirements defined in the latest edition of UFC 3-600-01, and/or equivalent Host Nation DIN 14675. The Fire Protection Engineer shall be directly involved in the design and review of all fire protection systems, life safety features, and applicable construction criteria for the project, and shall review the 100% design submittal, including all plans and specifications and certify in writing that the design complies with all applicable criteria,

including UFC 3-600-01 and relevant NFPA codes.

The design shall address the class of construction as defined by current applicable codes:

- Identification of occupancies, including rooms or areas used for storage of combustible or noncombustible materials or hazardous operations
- Type and scope of automatic sprinkler systems, special extinguishing systems, and fire alarm or evacuation alarm systems
- Location, size, and fire rating of required stair and shaft enclosures, door classifications, and door operation types
- Location and fire-resistance rating of fire walls, partitions, and fire doors
- Schedule of fire-resistance ratings for major building elements
- Requirements for electrical equipment, fixtures, ventilation, and related fire protection features for operations involving flammable liquids, gases, or dust
- Statements or notations concerning outside fire protection hydrant systems and any special water storage or pumping facilities required for deluge systems

2.6.4 Anti-Terrorism Force Protection (ATFP). The A&E shall incorporate all applicable Anti-Terrorism and Force Protection (AT/FP) requirements into the project design in accordance with DoD, Service-specific, and Host Nation criteria. All AT/FP features shall comply with the latest editions of UFC 4-010-01, UFC 4-020-01, UFC 4-022-01, and all other relevant UFCs, DoD Instructions, and installation-specific guidance. The A&E shall evaluate the project site, building configuration, and mission requirements to determine all necessary AT/FP measures, including standoff distances and site planning, hardened construction, blast-resistant design, vehicle barriers, relocation or protection of critical assets, structural and architectural AT/FP measures, mechanical and electrical system protection, and electronic security integration. The A&E shall provide all required AT/FP documentation with each design submittal, including the AT/FP Basis of Design, design analyses and calculations, drawings and details, blast and progressive collapse calculations (as applicable), an AT/FP compliance matrix, and documentation of coordination with Host Nation requirements. All submittals shall clearly demonstrate compliance with DoD minimum standards. The A&E shall coordinate all AT/FP design elements with the Installation Antiterrorism Officer (ATO), Security Forces, and other designated Government stakeholders, and shall document and submit for Government approval any AT/FP decisions requiring interpretation or deviation.

2.6.5 Secure Space Requirements. Design works involving secure spaces and ICD 705, UFC 4-010-05 requirements the A&E shall have U.S. citizens with the required clearance. The A&E shall have in staff, or subcontract, the services of qualified Construction Site Security Planning (CSSP) personnel to meet the requirements for security processes, notification, coordination, documentation, design, construction, and accreditation in accordance with

ICD 705 and applicable UFCs.

The CSSP services shall provide a qualified Site Security Manager (SSM) and an ICD 705 Subject Matter Expert (SME) during the design process through the award of the Construction Contract. The CSSP services shall develop and revise as necessary during the design period a separate Construction Security Plan (CSP). The CSSP services shall develop the initial draft after the design project is awarded. When finalizing the CSSP subcontractor, the SSM shall conduct an in-person or virtual stakeholders meeting at Ramstein AB, Germany with the owner of the secure space.

As the design progresses, the CSSP services shall update the CSP and complete annexes as required. The SSM and SME shall participate throughout the design phases (pre-design, initial site visit, concept/intermediate design at 35%–65%, Final Design at 95%, and Corrected Final Design at 100%). The SSM shall perform each design review and attend design review meetings, either virtually or in person as necessary. The CSSP services shall revise the CSP and create or update annexes as appropriate. This includes coordinating the site layout for the Secure Storage Area (SSA) and Access Control Facility (ACF) with the designer during design reviews and providing input on the design of the SSA and ACF based on site requirements.

The CSSP services shall develop the Construction Surveillance Technician/Cleared American Guard (CST/CAG) Support Plan in accordance with the A&E designs. After each design review, the SSM shall implement required changes to the CSP, Staffing Matrix, and DD Form 254 in accordance with the A&E design and the procurement plan. The CSSP services shall provide all required documentation and deliverables. The contractor's finalized CSP from the design process will be used for the separate construction/renovation process for the project requirement and for accreditation of the secure spaces.

- 2.6.6 Cybersecurity. The A&E shall provide qualified cybersecurity expertise either through in-house personnel or through subcontracted cybersecurity services. The Cybersecurity Designer shall be a recognized expert in FRCS cybersecurity and OT environments and shall be responsible for all tasks and objectives required by applicable laws, codes, standards, regulations, and this guideline. All design work shall comply with UFC 4-010-06, UFGS 25 05 11, DAFGM 2024-32-01, the IEC 62443 Series, the ISO/IEC 27000 Series, and all applicable Host Nation requirements.